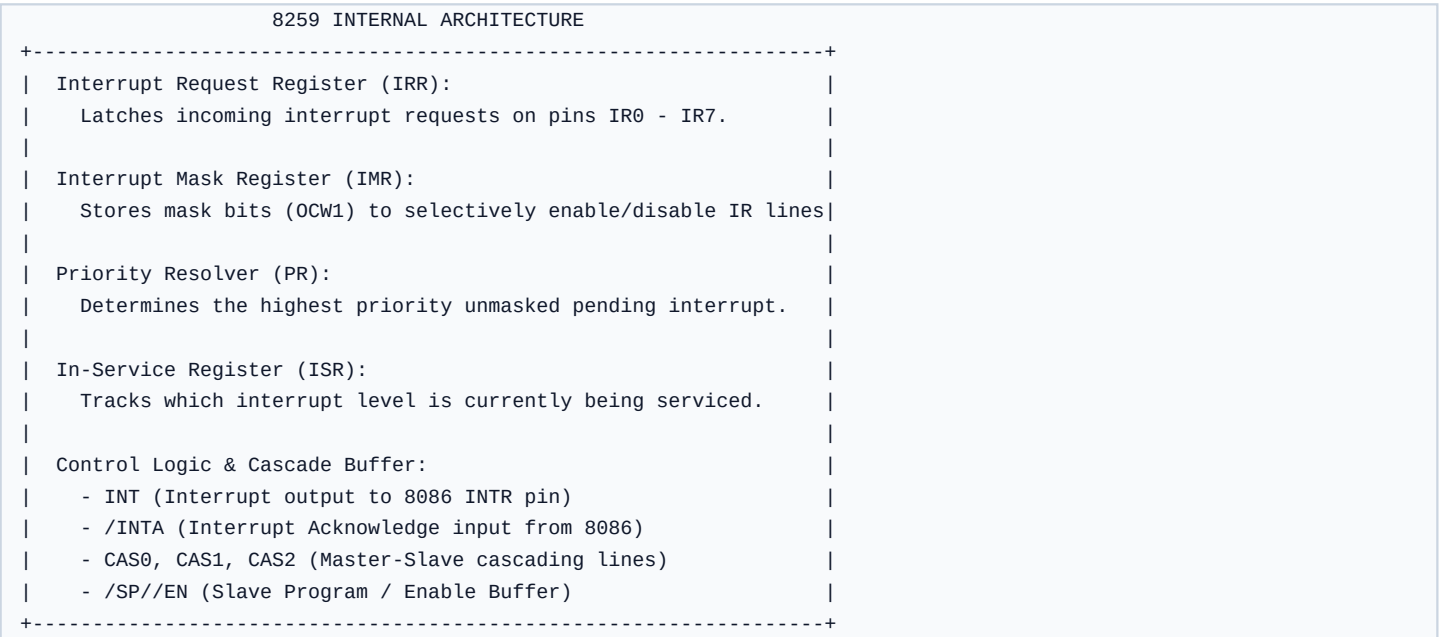


8259 Programmable Interrupt Controller (PIC) -- MPCA Module 4

Definition: The Intel 8259A (PIC) is an 8-channel programmable interrupt controller designed to manage, prioritize, and translate multiple hardware interrupt requests (IR_0 - IR_7) into 8086 vector interrupt instructions.

1. Detailed Technical Explanation

Internal Architecture & Block Diagram of 8259



2. 8259 Interrupt Processing Sequence

1. Peripheral asserts an interrupt line (IR_0 - IR_7).

2. IR

3. Priority Modes in 8259

1. Fully Nested Mode (Default): IR_0 has the highest priority and IR_7 has the lowest priority. Lower priority interrupts are inhibited while a higher priority ISR is running.

2. Au

4. Must-Write Points for Exams

- * IRR stores requested interrupts, ISR stores currently executing interrupt, IMR masks unwanted interrupts.

* Tw

5. Quick Recall Flow

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IR0-IR7 -> IRR -> Priority Resolver -> INT to 8086 -> 1st /INTA (Set ISR) -> 2nd /INTA (Send Vector Byte) -> Execute  
ISR -> EOI Command
```